

Final maths problem sheet – summer 2026 leaf finalist cohort

Level $x = x$ marks

No calculators

Level 1

1. Last year, an earthworm from Wigan named Dave wriggled into the record books as the largest found in the UK. Dave was 40 cm long and had a mass of 26 g. What was Dave's mass per unit length?

2. – *you must show your working!*

(i) Sketch the curve $y = \frac{1}{x-1}$.

(ii) Sketch the curve $y = \frac{x}{x-1}$.

Level 2

A list of five two-digit positive integers is written in increasing order on a blackboard. Each of the five integers is a multiple of 3, and each digit 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 appears exactly once on the blackboard. In how many ways can this be done? *Note that a two-digit number cannot begin with the digit 0.*

Level 3

- (i) Show that

$$\cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

and find a similar expression for $\sin 15^\circ$.

- (ii) Show that $\cos \alpha$ is a root of the equation

$$4x^3 - 3x - \cos 3\alpha = 0,$$

and find the other two roots in terms of $\cos \alpha$ and $\sin \alpha$.

- (iii) Use parts (i) and (ii) to solve the equation $y^3 - 3y - \sqrt{2} = 0$, giving your answers in surd form.

Level 4

Ada the ant starts at a point O on a plane. At the start of each minute she chooses North, South, East or West, and marches 1 metre in that direction. At the end of 2018 minutes she finds herself back at O . Let n be the number of possible journeys which she could have made. What is the highest power of 10 which divides n ?

Level 5

5. This question requires full written explanations.

The positive integers from 1 to N are equally spaced around a circle in such a way that the sum of any two neighbouring numbers is equal to the sum of the two numbers diametrically opposite them.

- (a) Figure 5 shows an example with $N = 6$. Number 1 is diametrically opposite number 4 and number 2 is diametrically opposite a .

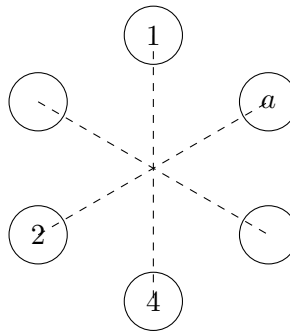


Figure 5

Copy and complete the diagram to show one possible way of arranging numbers 1 to 6. No explanation is needed in this part.

In the rest of this question, $N = 222$.

- (b) Show that the difference between any pair of diametrically opposite numbers is the same. You may find it helpful to refer to Figure 6 in your explanation.

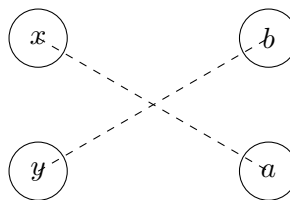


Figure 6

- (c) Suppose that there is an arrangement in which number 41 is opposite number 1. By considering which numbers would need to be placed opposite 40, 120 and 200, show that such an arrangement is not possible.

- (d) Find all possible numbers which could be placed opposite number 1. You need to explain why those are the only possibilities. In each case, you also need to explain why it is possible to arrange the remaining numbers.

Extra Extra Challenge

Completely optional; worth 10 additional points, separate from your total score. Significantly harder than the rest (approaching undergraduate level) — included purely for fun.

A batch of N USB sticks is to be used on a network. Each stick has the same unknown probability p of being infected with a virus. Each stick is infected, or not, independently of the others.

The network manager decides on an integer value of T with $0 \leq T < N$. If $T = 0$ no testing takes place and the N sticks are used on the network, but if $T > 0$ the batch is subject to the following procedure.

- Each of T sticks, chosen at random from the batch, undergoes a test during which it is destroyed.
- If any of these T sticks is infected, all the remaining $N - T$ sticks are destroyed.
- If none of the T sticks is infected, the remaining $N - T$ sticks are used on the network.

If any stick used on the network is infected, the network has to be disinfected at a cost of $\mathcal{L}D$, where $D > 0$. If no stick used on the network is infected, there is a gain of $\mathcal{L}1$ for each of the $N - T$ sticks. There is no cost to testing or destroying a stick.

(i) Find an expression in terms of N , T , D and q , where $q = 1 - p$, for the expected net loss.

(ii) Let $\alpha = \frac{DT}{N(N - T + D)}$. Show that $0 \leq \alpha < 1$.

Show that, for fixed values of N , D and T , the greatest value of the expected net loss occurs when q satisfies the equation $q^{N-T} = \alpha$.

Show further that this greatest value is

$$\mathcal{L} \frac{D(N - T)\alpha^k}{N}, \quad \text{where } k = \frac{T}{N - T}.$$

(iii) For fixed values of N and D , show that there is some $\beta > 0$ so that for all $p < \beta$, the expression for the expected loss found in part (i) is an increasing function of T . Deduce that, for small enough values of p , testing no sticks minimises the expected net loss.